Using Epistemic Observability in Agentic Systems for Combating Hallucinations

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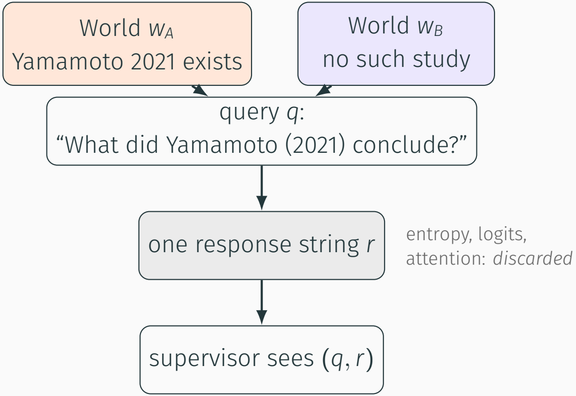
Which ten percent do you check?

- Some fraction of every model's output is fabricated: fluent, confident, wrong. Fabrication is provably inevitable for LLMs over computable functions (Xu et al. 2024). It is the expected behaviour of coherence optimisation, not a malfunction.
- Verification is not free. A citation lookup costs latency. A judge model costs compute. A human costs both, and costs the automation.
- So every deployment already allocates a budget: how much to verify, in what order, using what signal to choose.

The systems question

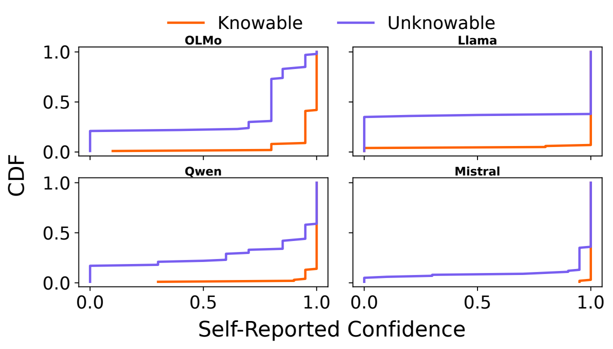
If the model exposed more than its text, what would that buy per unit of verification budget?

The observability gap



- Honesty asks for an answer in w_A and an abstention in w_B. The supervisor receives the same string in both.
- Under bounded supervision, grounded and fabricated states can be observationally equivalent through text, whatever the scale or training (Mason & Anand, arXiv 2026).
- The signals that could tell them apart were computed during generation and discarded at the interface. The supervisor pays to reconstruct them.

The cheapest fix: ask the model



"How confident are you?", asked after generation. Four instruction-tuned families, 200 queries each.

- Mean self-report 0.98 knowable, 0.73 unknowable.
- 43% of unknowable queries: a literal "100".
- The ordering exists (AUC 0.66–0.92) but the scale is compressed into its top end. A threshold that admits the correct answers admits the fabrications too.

Self-report is *testimony*: produced by the objective that produced the answer.

Testimony and telemetry

Testimony: the generated text. Optimised to satisfy the prompt, so it can mislead by construction.

Telemetry: byproducts of inference the model does not choose to emit: per-token entropy, log-probabilities, attention statistics. Under current training, changing them means changing the computation.

Epistemic observability: export the telemetry with the text so a downstream system can estimate fabrication risk. Not interpretability (why), not explainability (post-hoc rationale).

Tier	Exposes	Cost
0 Output	text only (today's APIs)	external checks
1 Confidence	logprobs, entropy	negligible
2 Representation	hidden states, probes	moderate
3 Mechanistic	circuits, information flow	high

A brainstorming tool needs Tier 0. Medical decision support may justify Tier 2 or 3. The hierarchy says what each level of investment buys.

A minimal tensor interface

What shape is wombat scat?

entropy: low mid high

`W ombat sc at is typically cylindrical or sausage -shaped . This form helps the sc at to be easily transported by wind or through the soil , aiding in dispers al and preventing it from being easily eaten by predators .`

Wombat scat is cube-shaped. The answer is a fabrication, and the model was uncertain exactly where it was inventing.

Returns: text, per-token entropy trace, attention summary.
No architectural change: the logits already exist.
Overhead 2–7% latency over generation.

Is not: a lie detector. Low entropy does not mean true. Entropy conflates epistemic uncertainty with aleatoric spread.
It *ranks* outputs for verification. It does not classify them.

Evaluation: return on verification investment

Queries: 100 knowable, 100 unknowable (nonexistent people, fictional syndromes, future events, fake citations). Labels by construction.

Models: OLMo-3 7B, Llama-3.1 8B, Qwen3 4B, Mistral 7B, all instruction-tuned. 800 responses.

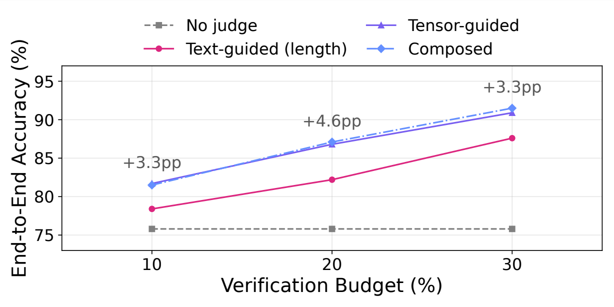
Budgets: the judge may verify 10%, 20%, or 30% of outputs. Confirmed fabrications become abstentions.

Four judges

1. No judge: raw output.
2. Text-guided: rank by response word count.
3. Tensor-guided: rank by mean entropy.
4. Composed: tensor ranking, plus an *oracle* for citation-like queries.

Verification is simulated (1,000 seeded trials): a selected fabrication is caught with $p = 0.938$, a selected error corrected with $p = 0.95$. The 0.938 is evaluator agreement with a blinded 80-item author review, used as a parameter, not a measured catch rate.

Tensor-guided verification wins at every budget



800 responses, four families, one budget.

+3.3 pp, **+4.6 pp**, **+3.3 pp** over the best text signal at 10, 20, and 30% verification.

Every family improves over no judge. At 10%: OLMo 69.5 → 75.1, Llama 82.5 → 88.1, Qwen 87.0 → 91.7, Mistral 64.0 → 72.0.

The entropy trace costs 2–7% latency and no change to the model.

Two signals, different items

Word count is a strong text signal. It ranks knowable against unknowable at AUC 0.85–0.97 per family. The tensor's budget advantage is not a better detector. It comes from *which items* each signal selects for verification.

And it is a signal training can move. Length and hedging are surface regularities; a model can be trained to change them without changing the computation beneath. The entropy trace moves only when the computation does.

Post-training does not erase it. Four public OLMo-3 checkpoints: base, SFT, DPO, RLVR.

- Knowable-query entropy falls $1.18 \rightarrow 0.18$ nats.
- Fabrications that survive stay high: $1.49 \rightarrow 0.98$.
- The separation *widens* through every stage.
- Refusals on fabrication prompts rise $0\% \rightarrow 56\%$.

Post-training improves the testimony and leaves the telemetry usable.

In a runtime: observe, then decide

The interface exposes *evidence*. A deployment policy chooses the *action*. Keeping those separate is the design.

Action ladder, increasing cost: abstain or flag $\to$ regenerate $\to$ tool verification $\to$ escalate to a higher tier $\to$ human review.

Three integration points

- **Serving layer**: telemetry exported with the text, amount set by the tier.
- **Composition boundary**: where the runtime intercepts tool and sub-agent calls, telemetry gates propagation. An unverified fabrication in step 3 of an agent is input to step 4.
- **Policy layer**: a budget allocator maps telemetry to actions from deployment-specific costs.

The ask

Signal access is a provider choice, and it is eroding. Log-probabilities were on early completion endpoints. They are absent from the newest reasoning models, the systems whose output is hardest to verify.

When only the provider can see the telemetry, only the provider can verify. Everyone downstream pays to reconstruct what the model already computed.

It works through an API, too. Top-5 log-probabilities from five serverless endpoints, 4B to 235B parameters including two mixture-of-experts models, separate knowable from unknowable at AUC 0.65–0.88.

Export the telemetry. It already exists. The interface is the only thing discarding it.

Summary

1. From text alone, a bounded observer cannot verify. Self-report is testimony and reads “100” either way.
2. Epistemic observability: export the telemetry inference already produces, in tiers priced by what they cost.
3. A Tier-1 entropy trace, 2–7% overhead, buys **+3.3 pp** to **+4.6 pp** end-to-end accuracy over the best text signal at every budget, across four families, and survives post-training.

artifact

Artifact:
`https://github.com/fsgeek/pacmi26-observability`
DOI 10.5281/zenodo.22162419 (v1.0.7)   Theory: arXiv, Mason & Anand 2026

Backup: the claim, precisely

Bounded supervisor. Observes $Obs(q, r, w) = (q, r, verify(r, w))$ when the verification cost is within budget B, and $(q, r, \emptyset)$ otherwise.

Hallucination regime. There is a plausible fabrication r_{fab} and worlds w_A (answerable), w_B (not) with verification cost above B in both, so $Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B)$.

Verification impossibility. Under that regime no bounded supervisor observing only text can return a verdict that is correct in both worlds; and no learning algorithm driven by that supervisor's reward can be guaranteed to converge to a policy that answers in w_A and abstains in w_B.

Stacking judges that consume only the previous judge's output cannot add information (observation monotonicity). Triage that re-inspects the original artifact, or any interface that exports state between layers, leaves the text-only class.

Mechanised in Lean; TLA+ models of both the text-only regime and the tensor interface, model-checked with TLC. In the tensor model *Verifiability* still fails at the plausible-lie case, where fluent fabrication and honest refusal are both low-entropy and coherent. That residue is the formal counterpart of AUC 0.87 rather than 1.0.

Backup: per-family and API results

Local, per family, own budget

Tensor over no judge at 10%: OLMo 69.5 $\rightarrow$ 75.1, Llama 82.5 $\rightarrow$ 88.1, Qwen 87.0 $\rightarrow$ 91.7, Mistral 64.0 $\rightarrow$ 72.0.

Tensor over length, averaged: +2.6 / +3.9 / +2.6 pp at 10/20/30%.

Mistral at 30%: length wins by 2.2 pp.

Detector AUC, knowable vs. unknowable

Entropy 0.87–0.92 per family.

Raw word count 0.85–0.97; beats entropy on Llama (0.949 vs 0.922) and Mistral (0.969 vs 0.905).

Cross-model rank correlation of entropy $\rho = 0.76$.

Serverless APIs, top-5 logprobs only

Five further endpoints, 4B–235B, including two mixture-of-experts (Llama-4 Maverick 128E, Qwen3-235B) and Gemma 3n.

Mean entropy AUC 0.65–0.88.

Cross-model $\rho = 0.36$; more architectural diversity and a top-5 approximation, not signal absence.

Peak entropy helps the MoE models; exploratory, not held-out.

Backup: what the paper does not claim

- That entropy is the better *detector*: length ranks at AUC 0.85–0.97 and beats entropy (0.87–0.92) on Llama and Mistral. On Mistral at 30% budget the length judge wins by 2.2 pp.
- That the composed judge is achievable: it reads the ground-truth label for citation queries, so its 91.5% is the headroom a perfect lookup would buy.
- That the post-training ceiling is 1.00: on the 48 probes it is; counting the true-but-implausible category gives $0.63 \to 0.81$–0.87. The widening is robust, the ceiling is not.
- That the tensor is unfakeable under adversarial training, that it survives composition in recursive pipelines, or that the design is optimal.
- Closed-weight models whose providers export no log-probabilities remain untested.

Backup: the confounds we know about

- **Length.** Unknowable prompts are longer (10.5 vs 7.0 words). Stratifying by query length leaves length at AUC 0.895 and entropy at 0.872. The confound is real and does not explain the signal.
- **Citations.** Flagged fabricated citations have *higher* entropy than the rest (0.575 vs 0.436). The composed judge is an oracle for that class, so its 91.5% is an upper bound.
- **Refusals.** Trained refusals are low-entropy. As a model gets more honest, honest refusals and confident facts share the same region of entropy space; the signal's power to catch *dishonesty* degrades. Private-context probes are excluded from the post-training diagnostic for this reason.
- **Labels.** 370 of 800 responses labelled programmatically, 430 by Qwen3-4B, itself a model under test; the labelling task differs from the generation task, and the blinded 80-item review is the check on it.